

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000206
Award Program	2018 Seeding Solutions: Addressing Challenges in Food and Agriculture Funding Opportunity
Project Title	Reducing food waste by re-shaping consumer behavior using data-informed, dynamic economic incentives
Grant Start Date	02/01/2019
Grant End Date	01/31/2023
Total Grant Budget	\$1,742,877.40
Total FFAR Budget	\$588,229.11
Matching Funder(s)	\$1,154,648.29
Final Report Date	05/01/2023
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district – where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Ithaca, NY, Congressional District 19

- 1.2. How many new jobs were created because of this grant funding?

0

- 1.3. How many jobs were maintained because of this grant funding?

2

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

A large proportion of food produced in the US is lost or wasted every year due to microbial spoilage. The development and implementation of innovative technical solutions that address microbial spoilage would substantially contribute to resolving the economic and environmental

problem represented by food waste and loss. This project focused on fluid milk as a model system, as dairy is among the top food groups lost or wasted, and fluid milk represents more than half of the dairy product waste and loss. The project took a systems approach by exploring interventions at each step in the dairy farm-to-fork continuum. Key tools that were developed during this project are five computer models that allowed for the evaluation and selection of interventions with the most cost-effective impact on product shelf-life and food waste reduction. Two of the developed models include a user-friendly interface and are available to the public online. Several different interventions at the (i) farm, (ii) processor, and (iii) consumer-level were evaluated. (i) Two farm-level interventions targeted udder health to reduce the spore counts in raw milk that are responsible for approximately one half of fluid milk spoilage; these interventions were identified as effective; however, they need to be a part of a more comprehensive control strategy for reducing spore loads and extending shelf-life. (ii) Bactofugation of raw milk and removal of spores at the processor-level showed a meaningful impact by providing an estimated two days of extended shelf-life. Prototypes of antimicrobial and nonfouling coatings were also developed that have the potential to be used as processor-level interventions to target post-pasteurization contamination by Gram-negative bacteria responsible for the other half of fluid milk spoilage. (iii) Three consumer-level intervention studies were conducted to determine consumer incentives for reducing food waste, including consumers' willingness to pay for shelf-life of fluid milk, consumers' acceptance of QR code technology, and impact of poor fluid milk experience on consumers' store choice. Results of the three studies showed that consumers are willing to pay a price premium of \$0.08 per day of extra shelf-life, and an additional \$1.13 if made aware of potential food waste reduction through appropriate messaging. Consumers are less willing to use QR codes even if it leads to purchasing product at a reduced price; the need to use QR codes reduces the willingness to pay by -\$0.86. A final online survey revealed that reduced shelf-life is associated with poor shopping experience which can lead to retailer losing consumer loyalty. This highlights the importance of extending the shelf-life of the product, not only for producers and processors, but also for the retailers. The results of this entire project were shared with the public, food industry, and scientific community at multiple occasions, including a two-day international virtual symposium on food waste. The predictive modeling tools developed and key findings of the project represent resources that have been used to support the dairy industry partners and continue research on the role of spoilage and consumer behavior on fluid milk loss and waste.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

There are several (i) key products and (ii) key findings that this project generated.

(i) The key products generated during this project are represented by five computer models, including three models that were developed to simulate and predict fluid milk spoilage due to outgrowth of two main groups of spoilage organisms, one model that was developed to

estimate “Cost-Benefit” of different interventions, and one model that was developed to optimize the use of interventions and based on processor size, volumes processed, and other characteristics determine what is the most cost-effective way of operating a fluid milk processing plant. The computer models we developed were specifically designed to be used as decision support tools by the dairy industry; however, these models also proved to be very useful as research tools in several different applications.

The use and benefit of these models as tools was demonstrated throughout this project to both direct our research and assist our industry partners in decision making. We used our “sporeformer” and “PPC” sub-models to collectively evaluate more than ten different interventions and determine in a digital environment which interventions are worth testing in the actual production or processing environment. For example, using the “sporeformer sub-model” we determined that each farm-level intervention targeting udder health would most likely not be cost-effective in reducing spoilage if not implemented together with other interventions as part of a larger and more comprehensive control strategy. On the other hand, using the same model to evaluate bactofugation we determined that this processor-level intervention would most likely be effective as a standalone strategy for extending shelf-life. This information was used to communicate to our industry partners the benefits of bactofugation and initiated an intervention study in the processing plant that confirmed the effectiveness and further validated the predictive model. A similar approach can also be taken with other interventions since the dairy industry is continuously looking for improvements that would bring them competitive advantage; for example, new technologies and new cleaning and sanitizing chemicals. Modeling tools we developed are an effective and low-cost solution for rapid screening of these improvements and selecting the ones with the most cost-effective potential. These models are also very flexible and based on the data available could be tailored to specific raw milk source(s), specific processing plant(s), or geographic location(s).

While dairy and specifically fluid milk were selected as the model system for this project similar approaches can be used for other industries, including produce, meat, and poultry. Computer models as tools are relevant to the entire food industry because they are applicable to any situation where reduced spoilage, food waste and costs, as well as improved production processes and increased profits are the desired goal.

Similarly, different government agencies that are involved in food security, food safety, and food quality, as well as government agencies and environmental organizations that are involved in protection of the environment, could benefit from the use of these tools in assuring a steady supply of safe, high-quality, and sustainable food that also minimizes the environmental impact by reducing food waste and loss. For example, food banks are by default dealing with foods with reduced shelf-life and potentially temperature abused foods; predictive models we developed could support them in evaluating the shelf-life of the products and help them prioritize distribution of product with shorter shelf-life.

These models can also be tailored to be used by the consumers and in this application they have the potential to also contribute to reduction of food waste; for example, the two models with user-friendly interfaces we developed were made available on-line and can be used by general public to assess the expected shelf-life of fluid milk based on the source of the product and conditions the product was exposed to; for example, the models could be used to

determine the remaining shelf-life of the product after failed grocery store delivery that resulted in product being left on the porch for extended time. Determining the shelf-life that was reduced due to this temperature abuse could make the difference between product being used or being discarded all together.

(ii) In addition, to key findings related to farm-level and processor-level interventions that target sporeformers in raw milk to extend the shelf-life of the final product, we also generated key information on potential future intervention that can reduce spoilage due to outgrowth of Gram-negative bacteria introduced into the final product as post-pasteurization contamination. We reviewed state-of-the-art antimicrobial and nonfouling technologies compatible with polyurethane and epoxy coatings suitable for Zone 2/3 regions, such as drains and equipment framework. Building on these available technologies we have developed a new method to synthesize a polyurethane coating with excellent coating and durability properties as well as introduced new low surface energy, nonfouling fluorinated block moieties. These prototypes of antimicrobial and nonfouling coatings can lead to multiple applications in the future either individually or combined, including as processor-level solutions that target post-pasteurization contamination.

Finally, this project generated some key findings related to consumer behavior; for example, we determined that consumer is willing to pay additional \$0.08 for each additional day of shelf-life which can represent the base for future introduction of dynamic shelf-life pricing. By setting the price based on the remaining shelf-life, the product with the longest remaining shelf-life would no longer be the most attractive option for the consumer and as a consequence the volume of product lost at retail level would be reduced. With dynamic shelf-life pricing in mind we also evaluated QR codes as technology that could facilitate communication of the dynamic information on shelf-life and price to the consumer. Our key findings are that for a low-cost product like fluid milk consumers are less willing to use QR codes even if it leads to purchasing product at a reduced price; the need to use QR codes was also shown to reduce consumers' willingness to pay.

Overall, the key products and findings of this project have important implications for the food industry since these can be used to inform and guide efforts to reduce food waste and loss throughout the entire food system.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

The main outcomes of this project are the (i) five models we developed. While two of the models were already used to make decisions related to our research and work of our industry partners, all five of the models we developed can lead to other **potential**

outcomes. Potential outcomes are also expected from our work on specific interventions, including (ii) farm-level, (iii) processor-level, and (iv) consumer level interventions.

For (i), all five of the predictive models were developed to assist the dairy industry in decision-making. Three of the models were developed to simulate and predict fluid milk spoilage due to outgrowth of two main groups of spoilage organisms. Although all three models were already extensively used in our research and work with the dairy industry, the flexibility in design and application of these models means that these models have even greater potential for use and development in the future. Depending on the available data, these models can be modified to fit almost any need related to prediction of shelf-life; for example, production, processing, and distribution conditions. For two of the models, we also developed a user-friendly interface and made them publicly available for general public to use them without the need for specific knowledge of computer modeling; for example, consumers can use it to evaluate the shelf-life of the product after temperature abuse. All the other models we developed (i.e., “Cost-Benefit” model and “Optimization” model) were shown to have useful functionalities that are equally flexible in design for any future applications; for example, to provide estimates of Cost and Benefit of different interventions, or to determine the most cost-effective way of operating a fluid milk processing plant.

For (ii), we performed farm-level intervention studies that targeted cow udder health, including cleaning of udders, training, and selection based on somatic cell counts and teat-ends conditions. The results of these studies showed that individually each of these interventions, although still statistically significant, had only a small impact on spore reduction and extension of shelf-life of the final product. Recommendations for dairy farms is to approach implementing improvements to farm hygiene practices from a holistic and multilevel approach because improvement at each level will result in small spore load reductions that incrementally, with each new level covered, will add-up to reductions that lead to substantial shelf-life extension.

For (iii), we evaluated and tested bactofugation as intervention for reduction of spores in raw milk. The results of this study showed that bactofugation can provide spore load reductions in raw milk to provide meaningful extension of shelf-life even when spore loads in the incoming raw milk are already low. Recommendation based on this result is that when processing raw milk with already low spore loads (e.g., raw milk from farms with good farm hygiene practices), the use of bactofugation can still be a beneficial in reducing the spore loads even further and adding additional days of shelf-life to the product; however, the benefits of bactofugation will be eliminated if used in a process flow where

post-pasteurization contamination is not under control.

For (iii), we also developed a prototype of antimicrobial and nonfouling coatings that have the potential to be used as processor-level solutions to target post-pasteurization contamination by Gram-negative bacteria. PPC is responsible for rapid spoilage of fluid milk, however, it is also sporadic and unpredictable even in processing plants where appropriate hygiene standards are maintained. Antimicrobial and nonfouling coatings can represent additional hurdle for these spoilage organisms; for example, by eliminating their ability to form biofilms in the processing environment which would result in reduced overall presence in the environment and their potential for contamination of the final product.

For (iv), we performed several consumer studies where we determined that a short shelf-life of fluid milk contributes to consumers' poor shopping experience and that if frequent or substantial can lead to retailer losing consumer loyalty. This importance to the consumer is also reflected in other results of our studies that show consumers are willing to pay a premium for milk with an extended shelf-life and they are also willing to pay additional premium if made aware of potential food waste reduction that they are contributing to. These results show great potential for introducing dynamic shelf-life pricing of fluid milk, including charging premiums or giving discounts based on remaining shelf-life. Dynamic pricing also has the potential to result in reduced loss of fluid milk at retail level since purchasing products with reduced shelf-life and price would represent a viable option for the consumer. While QR code technology was initially seen as a viable option for communicating the dynamic information on price and shelf-life, the results of our consumer studies indicated that consumers are reluctant to use QR codes and even reduces their willingness to pay. Based on this information the recommendation for the dairy industry is to find a different and maybe a more direct way of communicating the information on shelf-life and price; for example, beverage display cases with LCD doors where different dynamic information can be displayed without the need for consumers input.

- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

The main **intermediate outcome** of this project was the use of the predictive models we developed to assist specific dairy processors. We used two of the predictive models to collectively evaluate more than ten different intervention strategies. We also used these models to assist one of our industry collaborators in

evaluating bactofugation as a solution for reducing spore loads in raw milk and final product. With the help of this model, we were able to translate the results obtained during this work into expected days of extended shelf-life which our industry partner was able to use to make their own estimations of the cost-effectiveness of this intervention. This demonstrates that predictive models developed during this project are ready to be used by the dairy industry in different applications and demonstrates their usefulness as decision-making tools. In addition to the models that were developed during this project, investments and partnerships that were created as direct or indirect result of this project have been contributing to development of other prediction tools, including the ones for other dairy products (e.g., cheese and yoghurt) and other food products (e.g., produce) all of which are introducing improvements to U.S. agriculture and food systems by driving extension of shelf-life of different food products and reducing food waste and loss.

- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

There are two separate **end outcomes** that resulted from this project; (i) we trained young experts of tomorrow, and (ii) we made food industry in NY and U.S. more aware of digital solutions available to them and how these can contribute to food waste and loss reduction.

For (i), there were three separate post-docs, four graduate students, and two undergraduate students that were part of this project and received training in digital and systems approaches to sustainable food production. One post-doc received training in food safety and microbial food spoilage and was also assisted in professional development by offering support during job search and interviews for a faculty position. Two other post-docs involved in the project received training in modeling and optimization research. One graduate student, funded through the project completed her MS degree at the School of Hotel Administration at Cornell University; her thesis was entitled "Reducing Food Waste Through Quality Improvements and Information Disclosure in Food and Beverage Markets". Two graduate students and two undergraduate students that were involved in development of a polyurethane coating with nonfouling and antimicrobial properties learned technical skills in food science, polymer chemistry, microbiology, and food processing. While all these young professionals already proved them self as experts of tomorrow, some are already transferring the knowledge to other experts of tomorrow; for example, one of the post-docs is now a lecturer in supply chain, operations, and technology at Washington University in St. Louis.

For (ii), we managed to introduce to the dairy industry in NY and U.S. the concepts of digital dairy, Industry 4.0, computer modeling and how all of these relate and can be utilized in reducing food waste and food loss. Through direct interaction with individual dairy processors and also through our advisory council meetings, stakeholder meetings and final symposium on food waste, we were able to demonstrate the use of predictive computer models in specific situations as well as in broader context of digital solutions that food industry can benefit from.

Our efforts that are directly and indirectly related to this project were one of the first efforts in moving dairy processing in NY and nationally towards something that is collectively known today as Industry 4.0. Although this concept, that is based on proper data collection and data utilization, has already established itself in some parts of dairy production, the dairy processing and the general integration of the two is still lagging. The interest and further collaborations that were established with producers, processors, retailers and other companies as result of our work on this project are proof of the progress we made in changing the mindset on usefulness of different digital solutions.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Year 1 goals (Feb. 2019-Jan. 2020):

Stakeholder engagement:

- Assemble advisory council and hold first advisory council meeting (*Completed; Feb. 2020*).
- Hold annual open stakeholder input webinar (*Completed, Feb. 2020 & Mar. 2020. Two separate meetings with our industry partner to present the "sporeformer sub-model" and plan an implementation of intervention*).

Aim 1 (Prediction):

- Complete development of an initial fluid milk F2F spoilage model (*Completed, two manuscripts published*).
- Run initial sensitivity analyses (*Completed, two manuscripts published*).
- Create additional data needed for the spoilage model (*Completed, two manuscripts published*).

Year 2 goals (Feb. 2020-Jan. 2021):

Stakeholder engagement:

- Hold 2nd advisory council meeting (*Completed, Feb. 2021*).
- Hold annual open stakeholder input webinar (*Completed, Aug. 2020. Organized in collaboration with Cornell Institute for Food Systems*).

Aim 1 (Prediction):

- Perform model validation (*Completed, two manuscripts published*).
- Add cost benefit analyses to the simulation model (*Completed; manuscript submitted for review*).
- Continue creation of additional data needed for the spoilage model (*Completed, two manuscripts published*).

Aim 2 (Intervention):

- Evaluate possible interventions with stochastic spoilage model (*Completed, four manuscripts published*).
- Finalize interventions to be tested (*Completed, four manuscripts published*).
- Initiate at least one intervention study (*Completed, two manuscripts published, five submitted for review, and three in preparation*).

Year 3 goals (Feb. 2021-Jan. 2022):

Stakeholder engagement:

- Hold 3rd advisory council meeting (*Completed, Mar. 2022. Advisory council members attended our two-day virtual symposium on food loss and food waste in Mar. 2022; a document summarizing our work was sent to them after the symposium*).
- Hold annual open stakeholder input webinar (*Completed, Sep. 2021*).

Aim 1 (Prediction):

- Validate the spoilage model with independent data (*Completed, four manuscripts published and one manuscript in preparation*).
- Publish first paper (*Completed, six manuscripts published, five submitted for review, and three in preparation*).
- Make a model publicly available (*Completed, two manuscripts published, two models publicly available, and one model in the process of being made publicly available*).
- Hold at least one webinar that trains individuals to utilize and further develop the F2F spoilage model (*Completed, two video tutorials publicly available*).

Aim 2 (Intervention):

- Conduct intervention studies (*Completed, two manuscripts published, five submitted for review, and two in preparation*).

Aim 3 (Implementation):

- Estimate model of supply and demand for fluid milk (*Completed; manuscript submitted for review*).
- Develop prototype fluid milk package with dynamic shelf-life labeling systems (*Completed; one manuscript published, and one manuscript in review*).
- Initiate and complete consumer choice experiments (*Completed; manuscript submitted for review*).

Year 4 goals (Feb. 2022-Jan. 2023):

Stakeholder engagement:

- Hold 4th advisory council meeting (*Completed, Mar. 2023*).
- Hold annual open stakeholder input webinar (*Completed, Mar. 2022. Stakeholders attended our two-day virtual symposium on food waste and food loss*).
- Hold Global Summit on Reducing Food Waste (*Completed, Mar. 2022*).

Aim 1 (Prediction):

- Finalize user-friendly interface of the model (*Completed; two models publicly available, and one model in the process of being made publicly available*).
- Hold at least one workshop on predictive spoilage modeling (*Completed, two video tutorials publicly available*).

Aim 2 (Intervention):

- Finish intervention studies and data analyses and publish and disseminate results (*Completed, 39 information products created, including 6 published manuscripts, 5 submitted for review, and 3 in preparation*).

Aim 3 (Implementation):

- Test dynamic shelf-life labelling system in milk produced by Cornell dairy (*Completed, one manuscript published, and one manuscript submitted for review*).

2.5. Have any of the major goals/specific aims or milestones for the project

changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No changes.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Major project activities included work related to all three Aims of the project ("Prediction", "Intervention", "Implementation").

Under Aim 1 ("Prediction") we developed and revised five predictive models. We developed the (i) "PPC sub-model". We revised a previously developed model, (ii) "sporeformer sub-model". We integrated the two previous models into, (iii) "complete model" combining the functionality of both "PPC" and "sporeformer" sub-models. We also developed two additional models (iv) "Cost-Benefit sub-model", and (v) "optimization model".

For (i), the development of the "PPC sub-model" and validation were summarized and published in Journal of Dairy Science (<https://www.sciencedirect.com/science/article/pii/S0022030221010900>). The published paper includes results of the sensitivity analysis that we performed which indicate that in addition to storage temperature the maximum specific growth rate of individual spoilage allelic types is the key parameter that affects prediction of fluid milk spoilage due to PPC, providing a possible target for interventions; for example, a scenario we evaluated was using seek-and-destroy approach to eliminate specific allelic types with the highest growth rates from the processing environment. The developed model was shown as a useful tool to conduct this and other "What-If" scenarios which evaluated different interventions. Of four different general "What-If" scenarios that were evaluated, interventions that reduce the frequency of PPC occurrence were shown to be most effective in extending shelf-life; for example, implementing enhanced periodic equipment cleaning procedures on key pieces of equipment. Reducing the level of contamination when PPC occurs was also shown as effective if achieved reduction of the intervention is substantial; for example, reduction of 3 log units is expected to provide more than 7 days of additional shelf-life. A combination of both frequency and concentration reduction was evaluated in the scenario where antimicrobial and nonfouling coatings would be used on drains in the food processing facility; shelf-life was predicted to be extended by one day.

For (ii), the "sporeformer sub-model" that was initially developed and published has been

revised

(<https://www.sciencedirect.com/science/article/pii/S0022030218305253?via%3Dihub>). The revised version introduced two key innovations including (i) the ability to account for changing temperatures along the supply chain, and (ii) more optimized procedure of determining lag time of microbial growth. Different datasets were used for optimization and validation to allow assessment of (i) model adaptability to different time and temperature profiles and (ii) model generalizability to milk with different microbiological quality. Prior to the optimization and validation, the first model underwent a sanity check that followed the same steps we used for development of the original model using a historical dataset. The final optimized and validated model was used to perform sensitivity analysis which indicated that home storage temperature, facility-to-retail transportation temperature, and initial spore concentration were the three most influential factors for predicting milk spoilage by sporeformers. We also used this model to test two categories of "What-If" scenarios; first one was initial reduction of spore concentration (for example, by using bactofugation), and second one evaluated changes in temperature (for example, reducing fluctuation of temperature during different stages of storage and transport). Model predictions indicated that bactofugation with effectiveness of removing sporeformers from milk reported in the literature (2.2 log reduction) was the most effective strategy to reduce spoilage. The implementation of this strategy was predicted to reduce the percentage of spoiled milk from 55.3% to 36.3% on day 21 of storage and could delay the date, by which half of the milk lot produced was spoiled, from day 19 to day 28. Overall, the model provides a good framework for assessing the risk of fluid milk spoilage along the supply chain and evaluating the effectiveness of intervention strategies, including those that target storage temperatures at different supply chain stages. We are currently working on finalizing a manuscript where we will describe the improvements and benefits of the new simulation model; the manuscript will be submitted to Journal of Dairy Science for consideration and is expected to be published by the end of 2023.

For (iii), the "complete model" was developed, and it includes the revised version of the "sporeformer sub-model". This model allows for simulation and prediction of fluid milk spoilage caused by either post-pasteurization contamination (PPC) or sporeformers. The model separates the simulation iterations between the two sub-models based on pre-determined percentage of product that is expected to spoil by one or the other pathway; the assumption here is that even when both spoilage organism groups are present, spoilage due to PPC will always be the dominant and faster one. We are currently working on a "spoilage typing model" that will facilitate estimation of initial percentage of product that would spoil by one of two pathways that needs to be pre-determined for the "complete model". A "spoilage typing model" like this will allow for a rapid adjustment of initial spoilage parameters in the "complete model" based on shelf-life testing results; for example, testing results from specific dairy plant that "spoilage typing model" can translate into spoilage type and feed it to the "complete model" for shelf-life prediction and "What-If" scenarios evaluations. The development and use of "complete model" and "spoilage typing model" will be summarized in a manuscript that is expected to be sent to Journal of Dairy Science for consideration by the end of 2023.

For (iv), using the dataset we developed that combines (a) shelf-life data from the "PPC sub-model", (b) retail data on fluid milk prices, and (c) cost data for relevant intervention strategies, we developed a "Cost-Benefit sub-model". This model we developed evaluated five

different interventions at the processor-level that can address PPC and extend the product shelf-life: (1) seek and destroy, (2) improved preventative maintenance (PM) procedures, (3) increased periodic equipment cleaning, (4) use of a temperature sensor on the milk container to recognize when product was temperature abused, and (5) addition of antimicrobial and antifouling coatings to the drains. Based on our analysis, the economic benefit to the processor and environmental benefit of implementing each of the five interventions outweighed the costs. The intervention that provided both the greatest economic and environmental benefit was intervention (3), increased periodic equipment cleaning intervention. The estimated private benefit is between \$0.17 and \$0.20 per half-gallon milk, and the social benefit for this intervention is an additional \$0.12 and \$0.13 per half-gallon milk. The total cost of implementing this intervention is between \$25,667 and \$50,000 per processing facility, which equates to between \$0.001 and \$0.002 per half-gallon of milk. This work has been summarized in a manuscript that was submitted to Journal of Dairy Science for consideration and is expected to be published by the end of 2023.

For (v), we summarized and published in Frontiers in Sustainable Food Systems the development and use of a model with a different approach to supporting reduction of fluid milk spoilage (<https://www.frontiersin.org/article/10.3389/fsufs.2021.670029>). This mixed-integer linear programming (MILP) model was developed to optimize interventions and practices that reduce spoilage by sporeformers along the entire fluid milk supply chain. This “optimization model” considers all available interventions along the supply chain and based on processor size, volumes processed, and other characteristics determine what is the most cost-effective way of operating a fluid milk processing plant to maximize shelf-life, reduce cost, and reduce food waste. We considered two major groups of intervention strategies, including (i) introducing farm-level premium payments (i.e., bonuses) based on spore-forming bacteria counts in raw milk and (ii) investing in spore reduction technologies at the processing level of the fluid milk supply chain. We developed two versions of the mixed-integer linear programming model both of which assess improving milk shelf-life from the dairy processors' perspective. Our first version, imposed to a budgetary constraint, maximized milk's shelf-life to cater to consumers who prefer milk with a longer shelf-life. The second version minimized the budget required to perform operations to produce milk with a shelf-life of a certain length geared to certain customers. We generated case studies based on real-world data from multiple sources and used the models to obtain optimal solutions for different processor sizes. Results demonstrate that optimal combinations of interventions are dependent on dairy processors' production volume and quality of raw milk from different producers. With these two versions of the model, we showed that there are several ways of how modeling can be used as decision support tools that can aid individual processors in identifying the most cost-effective way of achieving a desired milk shelf-life.

In addition to developing these models we also developed user-friendly interfaces for some of the individual models, including “PPC sub-model” and “sporeformer sub-model”. The links to the model interfaces can be found on our “Digital Dairy” webpage that we launched in 2021 as part of this project (<https://cals.cornell.edu/digital-dairy>). The webpage provides access to our fluid milk predictive models, and also access to other useful information related to digital dairy, including other predictive modes we developed for cheese and fermented milks, recordings of talks from our virtual symposium “Towards a true systems approach to reducing food loss and waste from farm-to-table”, and scientific papers on digital dairy published by our

group. For each of the model interfaces, we also recorded a video tutorial on how to use the model and the interface which can also be found on our digital dairy webpage. Providing models in a user-friendly format and making them publicly available will facilitate the use of the models by the dairy industry, consumers, and other users. Once the “complete model” and “spoilage typing model” are finalized, we will also make the user interface and video tutorial for these models available on our webpage.

The work under Aim 2 (“Intervention”) included two intervention studies to reduce fluid milk spoilage due to sporeformers, including (vi) two Farm-level intervention studies that target cow udder health and (vii) a Processor level intervention study on use of Bactofugation. Our work under Aim 2 also included (viii) one Processor level intervention study to reduce fluid milk spoilage due to PPC and three Consumer level intervention studies to determine consumer incentives for reducing food waste, including (ix) consumer willingness to pay for shelf-life of fluid milk, (x) consumer acceptance of QR code technology to communicate precise spoilage information, and (xi) effects of poor fluid milk experience on store choice in different channels.

For (vi), we summarized and published how we evaluated a low-cost intervention that targets cow udder health by implementing washing towels with chlorine bleach and drying them completely, as well as training of milking parlor employees to focus on teat-end cleaning (<https://www.sciencedirect.com/science/article/pii/S0022030223000310?via%3Dihub>). In the first step of the study, we analyzed the costs of implementing this intervention. Our analysis determined that the cost to implement the intervention ranged between \$9.49 and \$13.35 per cow per year based on herd size, or between \$0.03 and \$0.05 per hundred-weight of raw milk, assuming a herd average 28,000 lbs. produced per cow per year. This analysis showed that proposed intervention is sufficiently low-cost to be applicable to any farm size and status. We used our previous data on spore reductions in raw milk when this approach is applied and predicted the expected extension of shelf-life of pasteurized milk using our “sporeformer sub-model”. The model predicted that 18.24% of half-gallon containers of pasteurized milk processed from raw milk receiving no intervention would exceed the Pasteurized Milk Ordinance (PMO) limit of 20,000 cfu/mL after 17 days of shelf-life while 16.99% of containers processed from raw milk receiving the intervention would reach this level by day 17 of shelf-life: this is a reduction of 1.25 per-centage points which corresponds to 6.85% reduction of product above the PMO limit or less than half of a day of extended shelf-life. In addition to this published study we also initiated a second study to further evaluate cow udder health as a potential target for spore interventions; we evaluated two aspects of cow udder health (a) Somatic Cell Count (SCC), and (b) teat-ends condition. For (a), we again used the “sporeformer sub-model” and data from our previous studies that included SCC and spore loads in raw milk from 92 dairy farms to predict the shelf-life of product made from raw milk from these farms. These results were compared to model predictions in two hypothetical intervention scenarios: (1) only raw milk from 10% of the farms with the lowest SCC (<100,000 cells/mL) is used in the production, and (2) raw milk from 10% of the farms with the highest SCC (>630,000 cells/mL) is eliminated and not used in production. The model predicted that (1) eliminating 10% of the milk with the highest SCC would not extend the shelf-life of the product while using (2) the 10% of milk with the lowest SCC would extend the shelf-life by 2 days. For (b), we performed a study that assessed spore loads in raw milk obtained from teats with two extremes of teat end condition; a score of 1 indicating teat-ends

that are in optimum condition, and a score 4 indicating teat-ends that show the highest level of damage. Although still significant difference, a minimal spore load difference was found in raw milk obtained from udders with different teat end conditions. The data from this study together with data on distribution of different teat end conditions and SCC we obtained from our partners from Cornell's School of Veterinary Medicine is currently being evaluated using our "sporeformer sub-model" and will be summarized in a manuscript that is expected to be sent to Journal of Dairy Science for consideration by the end of 2023. Final conclusion from the two studies on cow udder health is that these strategies have to be used as part of a broader and more comprehensive strategy that targets farm hygiene practices at multiple levels and this way a reduction of spore loads in raw milk can be achieved at a level that results in meaningful extension of shelf-life of the final product.

For (vii), a study that was summarized and published in Journal of Dairy Science was conducted with one of our dairy industry partners to evaluate the effectiveness of bactofugation in reducing spore loads in raw milk and the impact it can have on the shelf-life of the final pasteurized milk

(<https://www.sciencedirect.com/science/article/pii/S0022030222005719?via%3Dihub>). We collected samples from different stages of fluid milk processing with and without the use of bactofuge and we tested the samples for total bacterial count and different sporeformers including cold-tolerant sporeformers that are one of the main groups of sporeformers responsible for fluid milk spoilage. Results were evaluated to determine the count reductions during different stages of fluid milk processing and compare counts in fluid milk processed with and without bactofugation. On average, bactofugation reduced the total plate count by 1.81 ± 0.72 log cfu/mL and cold-tolerant sporeformers by 0.86 ± 0.59 log cfu/mL. We used the "sporeformer sub-model" and counts of cold-tolerant sporeformers in the final skim milk product to predict and compare the shelf-life of the product processed with and without bactofugation. The simulation model showed that bactofugation would extend the shelf-life of pasteurized milk by approximately two to three days depending on how shelf-life is defined and determined. The results of this study were presented to the dairy industry partner that was able to use the information to further evaluate the cost-effectiveness of using bactofuge in the processing flow. In addition to directly and indirectly helping fluid milk processors evaluate the benefits of using bactofugation as an intervention in their plants, this work also promoted the benefits of using mathematical modeling in decision making; for example, during the project we met with our industry partner to discuss the results and introduce the benefits of mathematical modeling.

For (viii), we summarized and published in ACS Food Science & Technology the state-of-the-art antimicrobial and nonfouling technologies compatible with polyurethane and epoxy coatings suitable for Zone 2/3 regions, such as drains and equipment framework; these non-food contact surfaces in food processing facilities are known harborage points and niches where spoilage organisms and pathogens can survive, grow and be introduced into the rest of the food processing environment, including food and food contact surfaces (<https://pubs.acs.org/doi/10.1021/acsfoodscitech.2c00148>). Building on these available technologies we have developed a new method to synthesize a polyurethane coating with excellent coating and durability properties as well as introduced new low surface energy, nonfouling fluorinated block moieties which together result in high molecular weight polymers that retain transparency and with it increase the potential application of these coatings. The

development and evaluation of these coatings was summarized in a manuscript that was submitted to the journal Coatings for consideration. In addition, we also synthesized several diol-derivatized quaternary ammonium compounds of varying carbon chain lengths and we determined the minimum inhibitory concentrations for several bacterial species including *Pseudomonas poae*. The work related to synthesis and characterization of five antimicrobial quaternary ammonium bromides (QAB) suitable for polymerization with common polymeric monomers to create antimicrobial polyurethanes and polyesters was also summarized in a manuscript that was sent to journal Food Microbiology for consideration. These coatings and compounds we developed were evaluated and developed for use as standalone coatings with either antimicrobial or nonfouling properties; however, we also synthesized a polyurethane and perfluoropolyether (PFPE) copolymer coating modified with QAB that combines the two properties. The combination of low surface tension from the PFPE and antimicrobial from the QAB produced a multifunctional polyurethane coating suitable for coating non-food contact surfaces in the production and processing environment that can short-term eliminate bacteria on contact and long-term create an inhospitable environment for bacterial growth. The development and evaluation of this coating was summarized in a manuscript that was submitted to journal Materials for consideration. One way these coatings could be used in the future are antimicrobial and nonfouling coatings on processing drains that are known to harbor large microbial loads and represent harborage points for number of spoilage and pathogenic bacteria. Using our "PPC sub-model" we evaluated this scenario and predicted that the combination of both reduced frequency of PPC and reduced concentration of contamination when PPC occurs as a result of using these coatings on drains would extend the self-life of pasteurized milk by one day. These predictions were also included in the publication that describes our "PPC sub-model" together with predictions on other "What-If" scenario (<https://www.sciencedirect.com/science/article/pii/S0022030221010900>).

For (ix), we performed a full consumer study using hypothetical conjoint choice experiment survey using an instrument we developed. We determined that consumers are willing to pay a premium for milk with an extended shelf-life. Price premium consumers are willing to pay for half-gallon container of milk was determined to be \$0.08 for each day of extra shelf-life and they are willing to pay additional \$1.13 if they are made aware of potential food waste reduction; for example, by appropriate messaging on the label. However, use of smart label, like a QR code, creates a disutility for the consumer which reduces the willingness to pay by - \$0.86. The study was summarized in a manuscript that was submitted to Journal of Dairy Science for consideration and is expected to be published by the end of 2023.

For (x), a study that was summarized and published in JDS Communications was performed in a student cafeteria to determine consumer acceptance of QR code technology (<https://www.sciencedirect.com/science/article/pii/S2666910222001211?via%3Dihub>). The study confirmed the results from the hypothetical conjoint choice experiment and showed that consumers prefer not to use QR codes even if the process of using them can lead to reduced price of the product; consumers purchased product with QR codes without scanning the QR code. Initially increased sales of milk with QR codes returned back to normal in the second month once QR codes appearing on the label were not perceived as novelty anymore.

For (xi), we performed an online survey to evaluate consumer perceptions of a poor fluid milk experience. The survey showed that almost half of the participants see poor fluid milk

experience as buying milk that is about to or already reached expiration date. Frequent poor experience with fluid milk was associated with a higher likelihood of switching stores and not being willing to recommend the store to others. Online grocery shopping was also associated with a higher likelihood of ordering milk when not needed; for example, through one-click reordering of established product basket. This study showed that the shelf-life of fluid milk is very much relevant to the retail part of the chain not just to producers and processors which might be another incentive for overall extension of shelf-life and consequently waste reduction.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

The final project proposal was developed to be scientifically rigorous, including basing the research project on appropriate data sources that include relevant peer-reviewed scientific information available at the time, and information provided by our food industry partners about the needs and challenges they are facing or anticipating. Once the final project was approved we ensured that our approach remains scientifically rigorous (i) by avoiding any deviations from the proposed project, (ii) by regularly updating and consulting our food industry partners inside and outside of the project Advisory Council on our progress and future plans, and finally (iii) by subjecting our approaches and findings to scientific scrutiny in the process of publishing our research in peer-reviewed scientific journals and presenting at scientific conferences.

- 2.8. List key stakeholders that could be served by results of this project.

The stakeholders that could be served by results of this project are numerous, including (a) dairy and other food industry, (b) consumers, (c) government agencies, and (d) environmental agencies. (a) The main stakeholders that could be served by results of this project are the dairy processors, dairy producers, as well as distributors and retailers that are working with fluid milk. Although we elected dairy, and specifically, fluid milk as a model system, the stakeholders that could be served are not limited to only dairy. Results obtained and tools developed during this project are relevant to the entire food industry because computer modeling can be applied to any situation where reduced spoilage, food waste and costs, as well as improved production processes and increased profits are the desired goal. (b) Consumers can benefit both directly and indirectly. Direct benefit can be utilizing predictive models, like the ones we made available online, to evaluate fluid milk and other products for a more accurate shelf-life; for example, home delivery of fluid milk that encountered unexpected temperature abuse of the product could be evaluated to determine how much the shelf-life was affected by the abuse. An indirect benefit for the consumer is simple access to fresh, safe, and high-quality milk and other food products that have a more predictable shelf-life. (c) Government agencies like FDA, USDA, EPA, and state agencies are involved on multiple levels, in food security, food safety, food quality and protection of the environment. The tools we developed and similar tools can already be used by these government agencies

to assist them in achieving these goals. (d) Environmental organizations have an interest in reducing waste and promoting sustainability. The use of computer modeling can help reduce the environmental impact of the food industry by reducing the amount of food loss and waste.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

As part of this project, all the results were disseminated to our industry Advisory Council in Feb. 2020, Feb. 2021, Mar. 2022, and Mar. 2023. We have also presented the concept of computer modeling and the benefits of it at the Stakeholder input webinar that was held on Sep. 2021; the webinar was attended by participants from industry, government, and academia. On two separate occasions, the same concepts were also specifically presented to the corporate quality management team of one of our dairy industry partners. In Mar. 2022 we organized a two-day virtual symposium titled "Towards a True Systems Approach to Reducing Food Loss and Waste from Farm-to-Table" where we presented on current challenges in reducing food loss and food waste, as well as on different approaches for resolving these using approaches developed during our project or other approaches our collaborators developed and presented on. Several round tables that were organized on different topics proved to be very beneficial to the participants, specifically one round table that covered the economic pitfalls of food waste and loss reduction strategies. This symposium was attended by 356 international participants and recordings of it were made available online to the public; currently the recordings have collectively 177 views.

As part of this work were also presented at several different conferences and meetings; for example, we participated with different contributions from this project at the International Association of Food Protection annual meeting three years in a row. In 2022, we participated with seven separate contributions in the form of oral presentation, poster, or symposia. In 2021, graduate student Samantha Lau presented a talk entitled "Application of Monte Carlo Simulation Model to Evaluate the Effectiveness of Different Interventions in Reducing the Spoilage of Pasteurized Fluid Milk Due to Post-Pasteurization Contamination" and in 2020 a talk entitled "Development of a Monte Carlo Simulation Model to Predict Pasteurized Fluid Milk Spoilage Due to Post-pasteurization Contamination". During the project we provided 13

additional contributions to other conferences and meetings in the form of articles, oral or poster presentations that discussed the work and results obtained during this project.

Finally, we also published six separate papers; (i) a paper on development and evaluation of our “optimization model”, (ii) a paper on development and evaluation of our “PPC sub-model”, (iii) a paper on effectiveness of bactofugation where our “sporeformer sub-model” was used to evaluate the impact on shelf-life, (iv) a paper on farm-level intervention targeting teat-ends where “sporeformer sub-model” was again used to evaluate the impact on shelf-life, (v) a paper on use of nonfouling and antimicrobial coatings in the food industry, and (vi) a paper on consumer perceptions of QR code technology. There are two other manuscripts that are currently under review at the Journal of Dairy Science and there are three additional manuscripts that are currently under development and will be sent out to peer-reviewed journals for consideration. Our work on “optimization model” together with the research team were also featured in an on-line article titled “Research provides blueprint for cost-effective, sustainable milk production” published by Phys.org (<https://phys.org/news/2021-07-blueprint-cost-effective-sustainable-production.html>).

In 2021, we also launched a new “Digital Dairy” webpage where we provide information on our current work related to development of different predictive models, including links to our models codes on GitHub. The webpage also provides links to user-friendly interface for two of our modes (i.e., “PPC sub-model” and “sporeformer sub-model”). Other useful information related to digital dairy that can be found on our webpage include other predictive modes we developed for cheese and fermented milks, recordings of talks from our virtual symposium on reducing food loss and waste, and scientific papers on digital dairy that our group published.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

While we experienced some delays due to COVID-19, there have been no challenges and delays during the project that have significantly impeded the research.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

The post-docs and the graduate students on the project participate in all project meetings and received mentoring and training that will prepare them for careers in digital and systems approaches to sustainable food production. Individual development plans were used to plan



coursework and support mentoring of all post-docs and graduate students. Graduate students and post-docs completed individual development plans through either Imagine PhD, myIDP Science Careers, or completed food science specific individual development plans. All post-docs, graduate and undergraduate students involved in this project have been given opportunities to present their findings at many levels: lab meetings, campus events, and national conferences, as well as in written technical reports. One graduate student, funded through the project in Year 3, completed her MS degree in August 2021 at the School of Hotel Administration at Cornell University. Her Master's Thesis entitled "Reducing Food Waste Through Quality Improvements and Information Disclosure in Food and Beverage Markets," has been published online through ProQuest (<https://www.proquest.com/docview/2581757112/3FDF4AB26BD04EF9PQ/1?accountid=10267>). Two graduate students and two undergraduate students that were involved in development of a polyurethane coating with nonfouling and antimicrobial properties learned technical skills in food science, polymer chemistry, microbiology, and food processing. One post-doc received training in food safety and microbial food spoilage and was also assisted in professional development by offering support during job search and interviews for a faculty position. Two other post-docs involved in the project received training in modeling and optimization research.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

3 post-docs
4 graduate students
2 undergraduate students
1 research associate
1 technician.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

6 Published scholarly publications
5 Scholarly publications in review
3 Scholarly publications in preparation
1 Master thesis
18 Conference proceedings
1 News article
2 Online event recordings

2 Tutorials

1 Website

3.2. Please provide a list of citations for the information products produced during the project.

1. *Published:* Evanowski, R. L., S. I. Murphy, M. Wiedmann, and N. Martin. 2023. Low-cost, on-farm intervention to reduce spores in bulk tank raw milk benefits producers, processors, and consumers. *J. Dairy Sci.* 106:1687-1694.
<https://www.sciencedirect.com/science/article/pii/S0022030223000310?via%3Dihub>
2. *Published:* Lau, S., A. Trmcic, N. Martin, M. Wiedmann, and S. I. Murphy. 2022. Development of a Monte Carlo simulation model to predict pasteurized fluid milk spoilage due to post-pasteurization contamination with Gram-negative bacteria. *J. Dairy Sci.* 105:1978-1998.
<https://www.sciencedirect.com/science/article/pii/S0022030221010900>
3. *Published:* Griep, E. R., A. Trmcic, C. Qian, and C. I. Moraru. 2022. Monte Carlo simulation model predicts bactofugation can extend shelf-life of pasteurized fluid milk, even when raw milk with low spore counts is used as the incoming ingredient. *J. Dairy Sci.* 105:9439-9449.
<https://www.sciencedirect.com/science/article/pii/S0022030222005719?via%3Dihub>
4. *Published:* Lau, S., M. Wiedmann, and A. Adalja. 2022. Short Communication: Consumer perceptions of QR code technology for enhanced fluid milk shelf-life information provision in a retail setting. *JDS Comm.* 3: 393-397.
<https://www.sciencedirect.com/science/article/pii/S2666910222001211?via%3Dihub>
5. *Published:* Rudlong, A. M., Y.T. Koga, and J. M. Goddard. 2022. Advances in Nonfouling and Antimicrobial Coatings: Perspectives for the Food Industry. *ACS Food Sci. Technol.* 2: 1401-1416.
<https://pubs.acs.org/doi/10.1021/acsfoodscitech.2c00148>
6. *Published:* Enayaty-Ahangar, F., S. I. Murphy, N. H. Martin, M. Wiedmann, and R. Ivanek. 2021. Optimizing pasteurized fluid milk shelf-life through microbial spoilage reduction. *Front. Sustain. Food Syst.* 5:670029.
<https://www.frontiersin.org/article/10.3389/fsufs.2021.670029>.
7. *Submitted to the journal for review:* Lau, S., M. Wiedmann, and A. Adalja. Economic and environmental analysis of processing plant interventions to reduce fluid milk waste. *Journal of Dairy Science*.
8. *Submitted to the journal for review:* Lau, S., M. Wiedmann, and A. Adalja. The effects of poor fluid milk experience on store choice and customer loyalty in different market channels. *Journal of Dairy Science*.

9. *Submitted to the journal for review:* Rudlong, A, and J. M. Goddard. Synthesis and characterization of antimicrobial hydrophobic polyurethane. Materials.
10. *Submitted to the journal for review:* Rudlong, A, and J. M. Goddard. Synthesis and Characterization of Hydrophobic and Low Surface Tension Polyurethane. Coatings.
11. *Submitted to the journal for review:* Rudlong, A, and J. M. Goddard. Synthesis and Minimum Inhibitory Concentration of Polymerizable Quaternary Ammonium Bromides. Food Microbiology.
12. *In preparation for submission to the journal for review:* Endara, P., M. Wiedmann, and A. Adalja. Consumer willingness to pay for shelf-life of high temperature-short-time (HTST)-pasteurized fluid milk: Implications for smart labeling and food waste reduction. Journal of Dairy Science.
13. *In preparation for submission to the journal for review:* Qian, C., S. I. Murphy, T. T. Lott, N. H. Martin, and M. Wiedmann. Prediction for milk spoilage along the supply chain due to psychrotolerant spore-formers using a supply-chain Monte Carlo simulation model. Journal of Dairy Science.
14. *In preparation for submission to the journal for review:* Trmcic, A., C. Qian, S. I. Murphy, Sunil, S., Evanowski, R. L., M. Wiedmann, and N. H. Martin. Using Teat Ends Score and Somatic Cell Count data as proxy to predict fluid milk spoilage using Monte Carlo simulation model. Journal of Dairy Science.
15. *Master's Thesis:* Endara, P. A. Reducing Food Waste Through Quality Improvements and Information Disclosure in the Food and Beverage Markets. Cornell University, ProQuest Dissertations Publishing, 2021. 28651675.
16. *Conference proceedings:* Trmcic, A., S. I. Murphy, N. H. Martin, and M. Wiedmann. Using computer modeling to evaluate suitability of intervention targets for reducing food spoilage: Example of raw milk Somatic Cell Count as target for reducing spores responsible for spoilage of fluid milk. IAFP Annual Meeting. Jul. 31-Aug. 3, 2022.
17. *Conference proceedings:* Trmcic, A., and S. I. Murphy. Computer Modeling-The Next Step in the Dairy Industry Evolution. IAFP Annual Meeting. Jul. 31-Aug. 3, 2022.
18. *Conference proceedings:* Murphy, S. I. Fluid Milk Spoilage Prediction Models. IAFP Annual Meeting. Jul. 31-Aug. 3, 2022.
19. *Conference proceedings:* Hubler S. R., Bulut E., Murphy S. I., Ivanek R. Quantitative microbial risk assessment of fluoroquinolone-resistant *Campylobacter jejuni* infections associated with unpasteurized milk consumption in the United States. 2022 National Veterinary Scholars Symposium, St. Paul, MN, Aug 5, 2022.
20. *Conference proceedings:* S. Lau, S. I. Murphy, M. Phillips, N. H. Martin, A. Trmcic and

- M. Wiedmann. Application of Monte Carlo Simulation Model to Evaluate the Effectiveness of Different Interventions in Reducing the Spoilage of Pasteurized Fluid Milk Due to Post-Pasteurization Contamination. IAFP Annual Meeting. Jul. 18-21, 2021.
21. *Conference proceedings:* Rudlong, Autumn; and Goddard, Julie. Synthesis and characterization of antimicrobial dihydroxy quaternary ammonium bromides. American Chemical Society National Meeting & Exposition. Atlanta, GA. August 22-26, 2021.
22. *Conference proceedings:* Goddard, J. M., Moderator; Gomes, CL; Zheng, J; Wiedmann, M, and Wu, F, Panelists. Digital Systems Approaches to Enhance Food Safety. Moderated Symposium at Institutes of Food Technologists Annual Meeting; July 2021
23. *Conference proceedings:* Doshna, N. A., Goddard, J.M. Antimicrobial active packaging synthesized by reactive extrusion of polypropylene and polylysine, a natural antimicrobial. Agricultural and Food Chemistry Division, American Chemical Society. Virtual, April 6, 2021.
24. *Conference proceedings:* Wiedmann, M. 2020. AI, Digital twins and modelling tools: A glimpse into the future of food safety, food quality, and sustainability. Southern California IFT monthly meeting. November 19, 2020.
25. *Conference proceedings:* Wiedmann, M. 2020. AI and Modelling Tools to Improve Food Safety. ILSI annual meeting, virtual. July 23, 2020.
26. *Conference proceedings:* Doshna, N. A., Goddard, J.M. IFTSA 20: Antimicrobial Active Packaging Synthesized through Reactive Extrusion. In: Undergraduate Research Competition. Institute of Food Technologists, Chicago, IL. July 22, 2020.
27. *Conference proceedings:* Doshna, N. A., Goddard, J.M. IFTSA 20: Antimicrobial Active Packaging Synthesized through Reactive Extrusion. In: Undergraduate Research Competition. Institute of Food Technologists, Chicago, IL. July 22, 2020.
28. *Conference proceedings:* Ivanek, R. Modeling of Food Waste. 2020 Cornell Symposium: Recent Advances in Food Systems (Food Waste: Challenges and Solutions). The Cornell Institute for Food Systems Industry Partnership Program (CIFS-IPP). Ithaca, NY. August 14, 2020.
29. *Conference proceedings:* Wiedmann, M. A Modeling Approach to Predicting Shelf-life to Reduce Food Waste. 2020 Cornell Symposium: Recent Advances in Food Systems (Food Waste: Challenges and Solutions). The Cornell Institute for Food Systems Industry Partnership Program (CIFS-IPP). Ithaca, NY. August 14, 2020.
30. *Conference proceedings:* Adalja, A. Food Waste Markets. 2020 Cornell Symposium: Recent Advances in Food Systems (Food Waste: Challenges and Solutions). The Cornell Institute for Food Systems Industry Partnership Program (CIFS-IPP). Ithaca, NY. August 14, 2020.



31. *Conference proceedings*: Enayaty-Ahangar, F., and S. I. Murphy. The challenges of microbial spoilage: Optimizing where dairy processors should invest. Cornell Dairy Center of Excellence Spring Seminar Series – 2020. Cornell University, Ithaca, NY. February 17, 2020.
 32. *Conference proceedings*: Enayaty-Ahangar, F. Optimization Model for Spoilage Reduction in the Milk Supply Chain. Biomedical & Biological Sciences (BBS) Ph.D. Recruitment Event Poster Session. Cornell University, Ithaca, NY. February 4, 2020.
 33. *Conference proceedings*: Enayaty-Ahangar, F. Optimization Model for Spoilage Reduction in the Milk Supply Chain. 2019 Doctoral Consortium on Computational Sustainability. Carnegie Mellon University, Pittsburgh, PA. October 19, 2019.
 34. *Other-Article*: S. Lau, and M. Wiedmann. Consumers embrace milk carton QR codes, may cut food waste. 2022. Cornell Chronicle. June 1, 2022.
<https://news.cornell.edu/stories/2022/06/consumers-embrace-milk-carton-qr-codes-may-cut-food-waste>
 35. *Online event recordings*: Towards a True Systems Approach to Reducing Food Loss and Waste from Farm-to-Table. Video on Demand. Cornell University.
<https://vod.video.cornell.edu/channel/Symposium%3A+Systems+Approach+to+Reducing+Food+Loss+and+Waste/251699423>
 36. *Online event recording*: Stakeholder webinar: Development of data-informed approaches to reducing food waste from primary production to consumer: Dairy as a model system. Cornell University.
<https://www.youtube.com/watch?v=bamp8yDiAUU>
 37. *Tutorial*: Digital tool for predictive modeling of fluid milk spoilage due to outgrowth of sporeformers. Cornell University.
https://www.youtube.com/watch?v=WXj-ZV_q_eQ&t=257s
 38. *Tutorial*: Digital tool for predictive modeling of fluid milk spoilage due to outgrowth of Gram-negative bacteria introduced in the milk through post-pasteurization contamination. Cornell University.
<https://www.youtube.com/watch?v=PMTp6Y-pnWs&t=348s>
 39. *Webpage*: Digital Dairy, Milk Quality Improvement Program & Food Safety Lab, College of Agriculture and Life Sciences, Cornell University.
<https://cals.cornell.edu/digital-dairy>
- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide

citation.

1. *Published:* Evanowski, R. L., S. I. Murphy, M. Wiedmann, and N. Martin. 2023. Low-cost, on-farm intervention to reduce spores in bulk tank raw milk benefits producers, processors, and consumers. *J. Dairy Sci.* 106:1687-1694.
<https://www.sciencedirect.com/science/article/pii/S0022030223000310?via%3Dihub>
 2. *Published:* Lau, S., A. Trmcic, N. Martin, M. Wiedmann, and S. I. Murphy. 2022. Development of a Monte Carlo simulation model to predict pasteurized fluid milk spoilage due to post-pasteurization contamination with Gram-negative bacteria. *J. Dairy Sci.* 105:1978-1998.
<https://www.sciencedirect.com/science/article/pii/S0022030221010900>
 3. *Published:* Griep, E. R., A. Trmcic, C. Qian, and C. I. Moraru. 2022. Monte Carlo simulation model predicts bacto-fugation can extend shelf-life of pasteurized fluid milk, even when raw milk with low spore counts is used as the incoming ingredient. *J. Dairy Sci.* 105:9439-9449.
<https://www.sciencedirect.com/science/article/pii/S0022030222005719?via%3Dihub>
 4. *Published:* Lau, S., M. Wiedmann, and A. Adalja. 2022. Short Communication: Consumer perceptions of QR code technology for enhanced fluid milk shelf-life information provision in a retail setting. *JDS Comm.* 3: 393-397.
<https://www.sciencedirect.com/science/article/pii/S2666910222001211?via%3Dihub>
 5. *Published:* Rudlong, A. M., Y.T. Koga, and J. M. Goddard. 2022. Advances in Nonfouling and Antimicrobial Coatings: Perspectives for the Food Industry. *ACS Food Sci. Technol.* 2: 1401-1416.
<https://pubs.acs.org/doi/10.1021/acsfoodscitech.2c00148>
 6. *Published:* Enayaty-Ahangar, F., S. I. Murphy, N. H. Martin, M. Wiedmann, and R. Ivanek. 2021. Optimizing pasteurized fluid milk shelf-life through microbial spoilage reduction. *Front. Sustain. Food Syst.* 5:670029. doi: 10.3389/fsufs.2021.670029.
<https://www.frontiersin.org/article/10.3389/fsufs.2021.670029>.
- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.
1. Digital Dairy, Milk Quality Improvement Program & Food Safety Lab, College of Agriculture and Life Sciences, Cornell University. Food Safety Laboratory and Milk Quality Improvement Program website includes a sub-page "Digital Dairy" where information about predictive models developed by our group and outside groups can be found separated by dairy commodity (e.g., Fluid Milk, Cheese, Yogurt). Research activities and products resulting from FFAR award are listed under section "Fluid Milk" together with a statement acknowledging FFAR financial support.
<https://cals.cornell.edu/digital-dairy>

2. GitHub website with "PPC sub-model" codes and input data.
https://github.com/FSL-MQIP/MC-2020/tree/master/Sam_FFAR_PPCmodel
3. GitHub website with "Optimization model" codes and input data.
<https://github.com/IvanekLab/Optimization-milk-shelf-life>
4. GitHub website with "Sporeformer model" codes and input data.
https://github.com/FSL-MQIP/MC_MilkSporeModel

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	0	0	
Copyrights	0	0	
Trademarks	0	0	
Inventions	0	0	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No.

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Models and input data for all published models are made publicly available through GitHub. The models and input data for the two currently published models and one that is in the process of being submitted to the journal for review can be found at these three links.
"PPC sub-model": https://github.com/FSL-MQIP/MC-2020/tree/master/Sam_FFAR_PPCmodel



"Optimization model": <https://github.com/IvanekLab/Optimization-milk-shelf-life>

"Sporeformer model": https://github.com/FSL-MQIP/MC_MilkSporeModel

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

- (i) Customer transaction data and QR code scan data for a pilot study conducted at Cornell Dairy Bar on QR code label acceptance + R code file to ingest data & analyze results.
- (ii) Analysis of value of shelf life using collected half gallon milk data and the cost benefit analysis of five interventions.
- (iii) Online survey data from 500 respondents for a discrete choice experiment to measure WTP for fluid milk shelf life + R code files to ingest data and conduct econometric analyses.
- (iv) Online survey data from 725 respondents for a consumer survey on the effects of a poor milk experience on store loyalty + R code files to ingest data & conduct econometric analyses.
- (v) The code for the base model, sensitivity analysis, what if analysis and validation of the PPC model with the results.
- (vi) Input data associated with 24 processors with different processed milk volumes + optimization codes in Python.

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

No.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

- (i) Customer transaction data and QR code scan data for a pilot study conducted at Cornell Dairy Bar on QR code label acceptance + R code file to ingest data & analyze results.
Data publicly available on GitHub: <https://github.com/sl2763/DairyBar>
Published in: <https://doi.org/10.3168/jdsc.2022-0256>
- (ii) Analysis of value of shelf life using collected half gallon milk data and the cost benefit analysis of five interventions.
Data publicly available on GitHub: <https://github.com/sl2763/CBAData>
Published in: To be published in 2023.



- (iii) Online survey data from 500 respondents for a discrete choice experiment to measure WTP for fluid milk shelf life + R code files to ingest data and conduct econometric analyses.
Data publicly available on GitHub: <https://github.com/aaronadalja/shelf-life-CBC>
Published in: To be published in 2023.
- (iv) Online survey data from 725 respondents for a consumer survey on the effects of a poor milk experience on store loyalty + R code files to ingest data & conduct econometric analyses.
Data publicly available on GitHub: <https://github.com/sl2763/ConsumerSurvey>
Published in: To be published in 2023.
- (v) The code for the base model, sensitivity analysis, what if analysis and validation of the PPC model with the results.
Data publicly available on GitHub: https://github.com/FSL-MQIP/MC-2020/tree/main/Sam_FFAR_PPCmodel
Published in: <https://doi.org/10.3168/jds.2021-21316>
- (vi) Input data associated with 24 processors with different processed milk volumes + optimization codes in Python.
Data publicly available on GitHub: <https://github.com/IvanekLab/Optimization-milk-shelf-life>
Published in: <https://www.frontiersin.org/articles/10.3389/fsufs.2021.670029/full>



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Martin Wiedmann

A handwritten signature in black ink, appearing to read "Martin Wiedmann", written over a light blue horizontal line.

PI Signature: _____ Date: 4/28/2023

Authorized Signing Official (ASO) Name: Jeffery D. Corbin

A handwritten signature in black ink, appearing to read "Jeffery D. Corbin", written over a light blue horizontal line.

ASO Signature: _____ Date: 4/28/2023